Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

August 7, 2019
WBL-19-039

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and NPF-96
NRC Docket Nos. 50-390 and 50-391

Subject: **Licensee Event Report 390/2019-002-00, Loss of Control Room Emergency Air Temperature Control System Due to Air Filter Failure**

This submittal provides Licensee Event Report (LER) 390/2019-002-00. This LER provides details concerning an incident where both trains of the Control Room Emergency Air Temperature Control System (CREATCS) were lost. This condition is being reported as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident in accordance with 10 CFR 50.73(a)(2)(v)(D).

There are no regulatory commitments contained in this letter. Please direct any questions concerning this matter to Tony Brown, WBN Licensing Manager, at (423) 365-7720.

Respectfully,

for Anthony L Williams

Anthony L. Williams IV
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant

NRC FORM 366 (04-2018)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020

LICENSEE EVENT REPORT (LER)

1. Facility Name	2. Docket Number	3. Page
Watts Bar Nuclear Plant, Unit 1	05000390	1 OF 5

4. Title

Loss of Control Room Emergency Air Temperature Control System Due to Air Filter Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
									Watts Bar Nuclear Plant, Unit 2	05000391
06	12	2019	2019	- 002	- 00	08	07	2019	NA	05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*			
1	[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)
	[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)
	[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)
	[ ] 20.2203(a)(2)(i)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)
10. Power Level	[ ] 20.2203(a)(2)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(v)(A)	[ ] 73.71(a)(4)
100	[ ] 20.2203(a)(2)(iii)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.71(a)(5)
	[ ] 20.2203(a)(2)(iv)	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(1)
	[ ] 20.2203(a)(2)(v)	[ ] 50.73(a)(2)(i)(A)	[X] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(i)
	[ ] 20.2203(a)(2)(vi)	[ ] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(vii)	[ ] 73.77(a)(2)(ii)
		[ ] 50.73(a)(2)(i)(C)	[ ] OTHER Specify in Abstract below or in NRC Form 366A	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Dean Baker, Licensing Engineer	(423) 452-4589

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	LE	FLT	IONEX	Y					

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
[ ] Yes (If yes, complete 15. Expected Submission Date) [X] No		N/A	N/A	N/A

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On June 12, 2019 at 0849 Eastern Daylight Time (EDT) a significant air leak on an inline air filter was identified. At 0908 EDT the leak on the filter was bypassed. This air leak impacted operation of the Control Room Emergency Air Temperature Control System (CREATCS) A Train. Two CREATCS trains are required to be operable in accordance with Technical Specification 3.7.11. At the time of this event, CREATCS B Train was out of service for planned maintenance.

The direct cause of this event is an air filter gasket leak. As corrective action, a different vendor approved gasket has been installed.

This condition is being reported as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident in accordance with 10 CFR 50.73(a)(2)(v)(D).

NARRATIVE

I. Plant Operating Conditions Before the Event

Both Watts Bar Nuclear Plant (WBN) Units 1 and 2 were at 100 percent rated thermal power (RTP).

II. Description of Event

A. Event Summary

On June 12, 2019 at 0849 Eastern Daylight Time (EDT) a significant air leak on an inline air filter {EIIS:FLT} was identified. At 0908 EDT the leak on the filter was bypassed. This air leak impacted operation of the Control Room Emergency Air Temperature Control System (CREATCS){EIIS:VI} A Train. Two CREATCS trains are required to be operable in accordance with Technical Specification (TS) Limiting Condition for Operation (LCO) 3.7.11. At the time of this event, the B Train of CREATCS was out of service for planned maintenance.

This event is being reported to the Nuclear Regulatory Commission (NRC) under 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

The B-B MCR chiller was out of service for maintenance when this event occurred.

C. Dates and approximate times of occurrences

Date	Time (EDT)	Event
6/03/19	N/A	Preventative Maintenance (PM) Work Order (WO) 119539837 performed to replace 0-FLTR-32-393 charcoal filter.
6/11/19	0400	Entered TS LCO 3.7.11 Condition A due to placing a clearance on the B-B Main Control Room (MCR) chiller in support of maintenance.
6/12/19	0325	Received 105B and 106B MCR HVAC SYSTEM A/B ABNORMAL alarms.
6/12/19	0332	Discovered 0-FCV-31-3, MCR Outside Air Isolation damper, had closed.
6/12/19	0345	Initiated B Train Control Room Isolation (CRI) in accordance with the Annunciator Response Instruction (ARI).
6/12/19	0844	MCR temperatures are above limit of 76 degrees Fahrenheit.

NARRATIVE

6/12/19	0849	Found significant air leak on 0-FLTR-32-393. Multiple TS entries made, including TS 3.0.3 for both train of CREATCS being inoperable.
6/12/19	0908	Bypassed 0-FLTR-32-393, isolating leak and restoring operability of affected plant equipment. Multiple TS exited.
6/12/19	1037	Reset B Train CRI.

D. Manufacturer and model number of each component that failed during the event

The leaking filter was manufactured by Ionex Research Corporation, gasket Part D-1440-4.

E. Other systems or secondary functions affected

The excessive filter leakage resulted in the closure of MCR ventilation dampers.

F. Method of discovery of each component or system failure or procedural error

The failure was identified by plant personnel investigating the closure of the MCR outside air damper.

G. Failure mode, mechanism, and effect of each failed component

A gasket failure occurred. The effect of the filter leakage was that downstream dampers closed.

H. Operator actions

Operations personnel responded in accordance with annunciator response instructions. Upon identifying the leaking air filter, the filter was promptly bypassed.

I. Automatically and manually initiated safety system responses

In response to low control room pressurization, operations personnel manually initiated a Train B CRI.

III. Cause of the Event

A. Cause of each component or system failure or personnel error

The direct cause of this event is an air filter housing gasket leak.

NARRATIVE

B. Cause(s) and circumstances for each human performance related root cause

No human performance related cause has been identified for this event.

IV. Analysis of the Event

The WBN MCR is provided with redundant trains of temperature control (CREATCS). In addition, redundant trains of control room emergency ventilation (CREV) are provided to support accident conditions to protect control room personnel from the release of airborne radioactive material. The air operated dampers that support operation of both of these systems are provided with safety related air from the auxiliary control air system (ACAS){EIIS:LE} which has redundant trains.

On June 12, 2019, at 0325, MCR HVAC abnormal alarms were received in the Main Control Room. Operations personnel responded in accordance with the appropriate ARI, and initiated a CRI Train B when the MCR Outside Air Isolation damper (0-FCV-31-3-A) was discovered closed and an attempt to re-open the damper failed. Investigation into the failed damper revealed that the gasket to the A train essential control air control building rad filter (0-FLTR-32-393) had failed which caused a loss of air pressure to multiple dampers serving CREATCS and CREVs. With the loss of air pressure, these dampers, including the MCR Outside Air Isolation damper and the modulating damper for the A train MCR cooling coil, went to the closed position. The closure of the modulating damper 0-FCO-31-82 effectively resulted in the loss of cooling from A train CREATCS

Un-recognized at the time the MCR Outside Air Isolation damper was discovered closed, was the loss of A train CREATCS. With the B-B MCR chiller out of service for maintenance, this resulted in both trains of CREATCS being inoperable, but was not recognized due to the slow heat up of the MCR. Operation of Train A of CREATCS was restored within TS LCO 3.0.3 requirements.

The A train of CREATCS was restored in less than 6 hours and MCR design temperature limits were not exceeded.

V. Assessment of Safety Consequences

Both Trains of MCR temperature control (CREATCS) were lost for less than six hours. The temperature rise in the MCR was limited to a maximum temperature of approximately 77 degrees Fahrenheit, with the design limit being 104 degrees Fahrenheit.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

There are no other systems that would perform the function of CREATCS.

NARRATIVE

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not applicable.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

For this event the CREATCS was returned to service in less than six hours.

VI. Corrective Actions

These events were entered into the Tennessee Valley Authority (TVA) Corrective Action Program and are being tracked under Condition Report (CR) 1524270.

A. Immediate Corrective Actions

Operations personnel responded in accordance with plant annunciator response instructions. Upon identifying the air leak, personnel promptly isolated and bypassed the leaking filter, restoring operability of Train A CREATCS.

B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future

As corrective action, a different vendor approved gasket has been installed.

VII. Previous Similar Events at the Same Site

An event with a similar cause was not identified.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.